

1. The level of measurement for the `hapmar` variable is _____

[1] "Happiness of marriage" Frequencies

`gss24$hapmar`

Type: Factor

	Freq	%	% Cum.
very happy	802	59.36	59.36
pretty happy	504	37.31	96.67
not too happy	45	3.33	100.00
Total	1351	100.00	100.00

- (a) dichotomous
 - (b) ordinal
 - (c) interval
 - (d) nominal
2. What does the pipe operator (`|>`) do in `dplyr`?
- (a) It filters rows based on a condition.
 - (b) It creates a new column in a data frame.
 - (c) It passes the result of one function into the next.
 - (d) It loads the `dplyr` package into memory.
3. Researchers found that couples report about 10 arguments per month on average. If a couple received a z-score of -2 for this measure, which of the following best describes the amount of arguments for this couple compared with the average?
- (a) This couple had fewer arguments than about 5% of couples.
 - (b) This couple had about 8 arguments per month.
 - (c) This couple had fewer arguments than about 98% of couples.
 - (d) This couple had fewer arguments than about 2% of couples.
4. What does the `::` operator do in R?
- (a) Imports all functions from a package
 - (b) Assigns a value to a variable
 - (c) Specifies a function from a particular package (without loading it)
 - (d) Comments out a line of code
5. Which of these datasets has a standard deviation of 0?
- (a) 0, 2, 4, 6, 8
 - (b) 2, 2, 5, 7, 7
 - (c) 4, 4, 4, 4, 4
 - (d) 1, 4, 7, 10, 15
6. What does the expression `5 != 3` evaluate to in R?
- (a) NA
 - (b) NULL
 - (c) TRUE
 - (d) FALSE
7. The variable `mntlhlth` measures how many days in the past 30 a respondent reported poor mental health. A respondent reports 5 poor mental health days. Based on the summary statistics for this variable, which of the following best describes their z-score?

freq	median	mean	sd
2279	0	4.53	7.93

- (a) The z-score is large and positive because 5 is much higher than the mean.
- (b) The z-score is undefined because the median is 0.
- (c) The z-score is negative because 5 is below the sd.
- (d) The z-score is close to zero because 5 is close to the mean.

8. Which function is used to load a package in R?

- (a) `library()`
- (b) `install.packages()`
- (c) `load()`
- (d) `source()`

9. Based on the table below, which statement best compares the relative frequency of being `not too happy` between men and women?

Cross-Tabulation, Column Proportions

`happy * sex`

Data Frame: `gss24`

	sex	male	female	Total
happy				
very happy		293 (20.2%)	386 (21.3%)	679 (20.8%)
pretty happy		830 (57.1%)	1054 (58.2%)	1884 (57.7%)
not too happy		330 (22.7%)	370 (20.4%)	700 (21.5%)
Total		1453 (100.0%)	1810 (100.0%)	3263 (100.0%)

- (a) A greater number of women reported being `not too happy` than men, so women have a higher relative frequency.
- (b) Men have a slightly higher relative frequency of being `not too happy` than women, even though fewer men reported it.
- (c) The frequency of being `not too happy` is equal for men and women.
- (d) Women have a slightly higher relative frequency because more women than men reported being `not too happy`.

10. What will happen if you try to use a function from a package that hasn't been loaded?

- (a) R will automatically load the package
- (b) You will get an error saying the function is not found
- (c) R will prompt you to install the package
- (d) The function will run normally

11. The U.S. General Social Survey asked people how many brothers and sisters they had. A researcher used R to produce the summary statistics of this variable (`sibs`) below. Based on this, you conclude the distribution for this variable is:

freq	median	mean	sd
3291	3	2.92	1.84

- (a) negatively skewed
- (b) positively skewed
- (c) nominal
- (d) symmetrical

12. When examining a frequency distribution, which of the following columns generally allows you to identify the median most quickly?
- (a) cumulative percentages
 - (b) percentages
 - (c) cumulative frequencies
 - (d) frequencies
13. Which RStudio pane is used to write and edit R scripts and Quarto documents?
- (a) Files
 - (b) Source
 - (c) Console
 - (d) Environment
14. What is Wheelan's primary message of Chapter 1 ("What's the Point?") in *Naked Statistics: Stripping the Dread from the Data*?
- (a) Statistics explain why relationships exist between variables
 - (b) Statistics provide honest answers to complex questions
 - (c) Statistical formulas are the most important part of understanding data
 - (d) Statistics simplify complex information to help us make better decisions
15. In a normally distributed set of test scores, a student scoring below the average will have a _____.
- (a) negative standard error
 - (b) negative z-score
 - (c) positive standard error
 - (d) positive z-score
16. Which command installs a new package in R?
- (a) `library()`
 - (b) `require()`
 - (c) `install.packages()`
 - (d) `load()`
17. In order to compare meaningfully across groups, a frequency should be converted into a/an _____.
- (a) number
 - (b) percentage
 - (c) standard deviation
 - (d) ordinal variable
18. Which of the following is a measure of the spread of a distribution?
- (a) standard deviation
 - (b) mean
 - (c) median
 - (d) standardized variable
19. What is the result of the following code?
- ```
gss24 |>
 filter(age > 20) |>
 select(age, childs)
```
- (a) The original `gss24` dataset is overwritten.
  - (b) A new column is added to `gss24`
  - (c) rows with `age > 20` are removed
  - (d) Only rows with `age > 20` and columns `age` and `childs` are returned.

20. What does Oscar Gandy suggest about the use of racial statistics?
- (a) Their use in legal proceedings increasingly lead to unfair policy outcomes
  - (b) They can both expose and perpetuate discrimination
  - (c) They are neutral tools most useful for academic research
  - (d) Their use in data journalism helps mitigate online polarization
21. If a statistics classes' final grades were normally distributed with an average of 81.5% and a standard deviation of 1.5, about what proportion of the students' final grades were between 80% and 83%?
- (a) 50%
  - (b) 62%
  - (c) 95%
  - (d) 68%

1. (a) ☐ (b) ☒ (c) ☐ (d) ☐
2. (a) ☐ (b) ☐ (c) ☒ (d) ☐
3. (a) ☐ (b) ☐ (c) ☒ (d) ☐
4. (a) ☐ (b) ☐ (c) ☒ (d) ☐
5. (a) ☐ (b) ☐ (c) ☒ (d) ☐
6. (a) ☐ (b) ☐ (c) ☒ (d) ☐
7. (a) ☐ (b) ☐ (c) ☐ (d) ☒
8. (a) ☒ (b) ☐ (c) ☐ (d) ☐
9. (a) ☐ (b) ☒ (c) ☐ (d) ☐
10. (a) ☐ (b) ☒ (c) ☐ (d) ☐
11. (a) ☒ (b) ☐ (c) ☐ (d) ☐
12. (a) ☒ (b) ☐ (c) ☐ (d) ☐
13. (a) ☐ (b) ☒ (c) ☐ (d) ☐
14. (a) ☐ (b) ☐ (c) ☐ (d) ☒
15. (a) ☐ (b) ☒ (c) ☐ (d) ☐
16. (a) ☐ (b) ☐ (c) ☒ (d) ☐
17. (a) ☐ (b) ☒ (c) ☐ (d) ☐
18. (a) ☒ (b) ☐ (c) ☐ (d) ☐
19. (a) ☐ (b) ☐ (c) ☐ (d) ☒
20. (a) ☐ (b) ☒ (c) ☐ (d) ☐
21. (a) ☐ (b) ☐ (c) ☐ (d) ☒